

Designing a Wearable AI Pendant with Solid-State Power and Ternary AI

Executive summary

The technically credible version of this product is **not** a continuously recording body camera. It is an **explicit-capture, episodic memory pendant**: front-facing camera, right-side physical button, dual microphones, strong local processing, and a very low-power always-on wake path that keeps the main camera and main compute asleep unless the user presses the button or a narrow trigger model fires. That framing aligns with the public power envelopes of always-on image sensors and wake processors such as HM01B0/HM0360 and WiseEye2, with higher-power burst capture from sensors such as OV2740 or IMX708-class assemblies. ¹

The central feasibility constraint is **energy storage, not vision silicon**. Publicly available solid-state microbatteries from TDK, Ilika, and ITEN are safe and compact, but their currently advertised capacities are in the **100–300 μAh class** or, for announced early ITEN parts, **150–250 μAh** , which is well suited to backup, sensing bursts, and tiny IoT nodes but not to an all-day camera pendant. Larger ceramic-cell claims from ProLogium and next-generation TDK materials show that the cell-level energy density required for a pendant is plausible in the future, but those claims do not yet translate into a widely sourceable, pendant-class packaged battery for mass production. ²

A **native-ternary hardware** product is still a research or niche-commercial path, not the prudent basis for a near-term wearable. Recent work on BitNet, bitnet.cpp, xTern, and ternary/MVL arithmetic shows that ternary neural networks and ternary kernels can be efficient, and small vendors now market research-oriented ternary processor boards. But the public toolchains of surveyed wearable-class chips remain overwhelmingly **INT8- or binary-centric**. The pragmatic path is therefore to make the product “ternary” first at the **model and algorithm layer**, while keeping the deployment path compatible with commodity edge silicon. ³

The best near-term architecture is a **three-domain system**: an always-on domain for wake sensing and button handling, a capture domain for short camera/audio bursts, and a richer inference domain for summarization or embedding generation. Among public options, the strongest balance for a research-oriented pendant is an **OV2740-class main camera**, optional **HM01B0/HM0360-class wake camera**, an **i.MX RT700-class** or **Apollo510-class** control/AI path, optional **NDP120-class** audio sidecar, and tightly duty-cycled BLE-first connectivity. ⁴

Thermally, a pendant is unforgiving. With a small enclosure and partial skin contact, sustained shell dissipation should typically stay below roughly **200–300 mW** if comfort is a goal; above that, burst-only operation, thermal spreading, and aggressive policy control become mandatory. Using the sourced public power numbers for sensors and NPUs, this means continuous live preview or frequent Wi-Fi upload is a poor fit for today's solid-state battery options, while episodic capture and metadata-first workflows are compatible. ⁵

Privacy and compliance should be treated as first-class product architecture. That means **hardware-visible capture indication, privacy-by-default local inference, secure boot, encrypted storage and transport, short retention windows**, and conformance planning across wireless, safety, battery transport, and consumer-IoT cybersecurity frameworks. In practice that brings in FCC equipment authorization, RED cybersecurity obligations in the EU, ETSI EN 303 645 baseline IoT controls, battery safety and transport requirements, and data-protection-by-design obligations for camera and biometric processing. ⁶

Assumptions and design targets

Because target price, exact dimensions, target market, and battery-life target were not specified, the report assumes the following design envelope:

- **Product class:** nonmedical consumer/prosumer wearable memory aid, not a diagnostic or industrial safety device.
- **Industrial design target:** pendant worn high on the chest, approximately **35 × 30 × 8 mm**, with a goal of **12–18 g** total mass.
- **User behavior:** mostly **button-driven capture**, with optional low-rate contextual triggers and short audio snippets rather than continuous livestream.
- **Connectivity model:** **BLE-first** with a companion phone; Wi-Fi only for sync, firmware update, or charging-dock offload; **no cellular** in v1.
- **Commercial target:** MSRP assumption of roughly **\$299 ± \$100**.
- **Battery target:** aspirational **8–12 hours worn time** with intermittent capture, which implies a strong need for aggressive duty-cycling and currently forces a research-vs-product tradeoff if all-solid-state is mandatory.

Under these assumptions, the design problem becomes a three-way optimization among **privacy, average power, and model flexibility**. If the actual intent is continuous lifelogging video, a much larger energy budget and a different regulatory posture would be required.

Product concept and reference architecture

The best product concept is an **AI notebook pendant with visual context**, not a surveillance device. The front-facing camera should serve three high-value use cases: explicit memory bookmarking by button press, contextual assistance such as “what object or screen am I looking at,” and low-friction incident capture for accessibility or safety workflows. That product concept is meaningfully different from a broad-field, always-on body camera because it lowers both average power and privacy risk.

For the hardware stack, the strongest public options today are the OV2740 from OmniVision ⁷, the HM01B0/HM0360/WiseEye2 family from Himax ⁸, and the IMX500/IMX708 ecosystem from Sony ⁹; main compute candidates are the i.MX RT700 from NXP Semiconductors ¹⁰, the STM32N6 from STMicroelectronics ¹¹, and the Apollo510 from Ambiq ¹²; companion radio or ultra-low-power control roles fit the nRF54L15 from Nordic Semiconductor ¹³, while always-on audio offload fits the NDP120 from Syntiant ¹⁴. Public documentation for these parts spans ultra-low-power always-on sensing, compact front-camera modules, and MCU-class NPUs, but none of them publicly exposes native ternary arithmetic as a first-class deployment target. ¹⁵

The recommended architectural pattern is **ternary in state structure**, even if not ternary in transistor logic: an **always-on domain**, a **burst capture domain**, and a **rich inference domain**. That structure is the cleanest way to reconcile low-power sensing, solid-state battery scarcity, explicit user control via the right-side button, and future experimentation with ternary neural networks.

```
flowchart LR
    BTN[Right-side physical button]
    MIC[Dual PDM microphones]
    SENS[IMU / prox / ALS / pack NTC]
    AON[Always-on domain\nbutton FSM + wake audio\noptional wake vision]
    CAM[Main front camera\nOV2740 or IMX708 class]
    MAIN[Main compute\nRT700 / STM32N6 / IMX500 path]
    STORE[QSPI NOR + optional eMMC]
    HAPTIC[LRA haptic + driver]
    RADIO[BLE default\noptional Wi-Fi]
    POWER[Solid-state pack\nPMIC + charger]

    BTN --> AON
    MIC --> AON
    SENS --> AON
    AON --> MAIN
    CAM --> MAIN
    MAIN --> STORE
    MAIN --> RADIO
    MAIN --> HAPTIC
    POWER --> AON
    POWER --> CAM
    POWER --> MAIN
```

Camera module and sensor options

Option	Publicly documented specs	Why it matters for a pendant	Main limitation	Evidence
OmniVision OV2740	1920×1080 active array, 1080p60, 90 mW active, 210 μA standby, 0.6 μA XSHUTDN, 5.5 × 5.5 × 3 mm module, LSM and ULP modes	Very strong main front camera for explicit burst capture ; unusually good combination of size and low active power for 1080p	Fixed-focus front-camera profile; no integrated AI	16

Option	Publicly documented specs	Why it matters for a pendant	Main limitation	Evidence
Sony IMX708 OEM assembly	11.9 MP, HDR, PDAF, 1080p50, 720p100, 75° diagonal standard lens or 120° wide lens variant, production listed through at least Jan 2030	Best if still images and autofocus matter; wide lens is attractive for chest-worn use	Higher image-data burden on host compute and storage than metadata-first options	17
Sony IMX500 intelligent sensor path	12 MP image sensor with on-sensor inferencing; designed to reduce bandwidth and power by keeping AI on-chip; outputs tensor metadata alongside video	Strong privacy architecture and lower host-memory pressure; attractive when metadata-first workflows dominate	Public deployment flow is constrained to supported single-image perception models and 8-bit conversion flow	18
Himax HM01B0	320 × 320 “always on” sensor, <2 mW at QVGA 30 fps	Excellent wake sensor for human-presence or pose triggers	Not a hero camera; image quality is for triggering, not rich capture	19
AR0144CS global-shutter path	1280H × 800V, 1/4-inch, 1 MP, global shutter, optimized for fast motion capture, low operational power, small form factor	Good for gesture, tracking, and motion-robust capture	More industrial/AR than consumer-photo oriented	20

A chest-worn pendant typically benefits from a **wider field of view** than a face-height clip. That makes an IMX708 wide-lens style option or a wide-angle OV2740-class integration more appropriate than a narrow smartphone selfie field, unless the product is clipped high on clothing. The tradeoff is more bystander capture and greater need for obvious recording cues. 21

Compute options and low-power AI accelerators

Option	Public capability	Public toolchain or deployment constraint	Ternary readiness	Best role	Evidence
Himax WiseEye2 HX6538	50 GOPS, 2.5 MB memory, 400 MHz Cortex-M55 + 400 MHz Ethos-U55 microNPU; public example shows VGA/QVGA person detection at 1 FPS with 1.2 mW average	TensorFlow Lite Micro and heterogeneous wake pipeline	Medium for research, because it is programmable and ultra-low-power, but not native ternary	Always-on visual wake-up	22
NXP i.MX RT700	Up to five cores, 7.5 MB SRAM, M33 + HiFi DSPs + Neutron N3-64 NPU; public NPU examples show up to 169× speedup and up to 83× lower energy per inference vs M33-only	eIQ Toolkit and Neutron conversion flow	Medium; good for custom low-bit experiments on CPU/DSP, but NPU flow is conventional	Main burst-inference MCU for a non-Linux pendant	23
ST STM32N6	Cortex-M55 at 800 MHz, 600 GOPS NPU, 4.2 MB embedded RAM; ST states about 3 TOPS/W for the NPU	STM32Cube.AI / ST Edge AI tools	Low-to-medium; excellent INT8 deployment, limited public ternary story	Higher-performance vision bursts where thermal budget permits	24

Option	Public capability	Public toolchain or deployment constraint	Ternary readiness	Best role	Evidence
Ambiq Apollo510 / Apollo510B	Cortex-M55, 300× more inference throughput per joule claim vs prior generation; Apollo510B adds BLE 5.4, 3.75 MB RAM, 4 MB NVM	neuralSPOT, TFLM, heliaRT/ heliaAOT	Medium for lightweight custom kernels; low for native ternary silicon ambitions	Control plane, light AI, radio integration, audio tasks	25
Sony IMX500 path	Vision AI on sensor with metadata output; reduces need for external processor and memory	IMX500 conversion and packaging flow; public docs indicate 8-bit quantized Keras/ ONNX deployment and supported single-image models	Low for ternary experimentation because the toolchain is tightly constrained	Metadata-first privacy-centric product	26
Syntiant NDP120	Always-on audio and sensor processor; MLPerf Tiny v0.7 results cite 35.29 µJ/ inference at a low-energy operating point	Vendor runtime centered on always-on audio/ sensor models	Low for general ternary compute, high as an audio wake sidecar	Voice activity detection, keyword spotting, sensor wake	27

Interpretation of “ternary readiness.” No surveyed wearable-class option publicly advertises native ternary MAC units, trit-addressable SRAM, or a balanced-ternary deployment toolchain. In this report, “ternary readiness” therefore means **programmability and low-bit execution flexibility**, not true MVL silicon. That is an inference from the public docs of Sony, Himax, NXP, ST, Ambiq, and Syntiant. 28

The memory architecture should be explicitly hierarchical. A practical pendant should use **on-chip SRAM for wake paths and fast inference, 16–64 MB QSPI NOR** for firmware and small models, and **8–32 GB eMMC** only if the product truly stores local images or audio instead of transient embeddings or encrypted snippets. RT700 already provides 7.5 MB SRAM; Apollo510B provides 3.75 MB RAM and 4 MB NVM;

nRF54L15 provides 1.5 MB NVM and 256 KB RAM; IMX500 shifts some of the bandwidth/storage burden by doing inference at the sensor and producing metadata. ²⁹

For **sensors**, the minimum practical set is a 6-axis IMU, proximity/ambient-light sensing near the camera, a battery NTC, and a dock/lanyard detect sensor. For **audio**, two PDM MEMS microphones are preferable: MP34DT05-class parts are compact and low-power, while IM69D130-class parts trade more acoustic performance for more demanding audio front ends. For **haptics**, a closed-loop LRA driver such as DRV2605L-class hardware is far preferable to a speaker beep because it preserves privacy and supports silent state confirmation. For **wireless**, BLE should be the default transport; Wi-Fi should stay off except during dock sync, firmware update, or explicit user action. ³⁰

The **right-side physical button** should be wired into the always-on domain, not only the application MCU. The correct behavior is: **short press** for explicit capture/bookmark, **long press** for privacy hold or emergency action, **double press** for mode switching. The button should also wake the PMIC or always-on controller directly so that a capture gesture still works from deep sleep.

For **connectors**, the most mechanically coherent consumer design is a **rear pogo or magnetic charge interface** plus hidden factory pads, not an exposed USB-C port. A direct USB-C opening improves universal charging and desktop serviceability, but it costs z-height, complicates IP sealing, and adds connector wear in a swinging pendant. The EU common-charger framework strongly reinforces USB-C across many handheld categories, but a pendant is still more constrained by **mechanical sealing and comfort** than by policy. ³¹

Battery and thermal feasibility

Publicly visible solid-state battery options relevant to this project come from Ilika ³², TDK ³³, ITEN ³⁴, and ProLogium ³⁵. Ilika's Stereax line is licensed for manufacture through Cirtex Medical ³⁶, TDK's CeraCharge is a reflowable oxide-based SMD microbattery, ITEN is publicly launching industrial-scale production of its Powency solid-state ceramic family, and ProLogium is demonstrating much larger cell-level densities and fast charging in ceramic batteries. The important distinction is that the first three are **microbattery-class**, while ProLogium's public performance is **larger-format and not pendant-package ready**. ³⁷

Solid-state battery options

Supplier and product	Public status	Publicly stated capacity or density	Form factor	Charging / safety note	Suitability for pendant	Evidence
TDK CeraCharge	Current SMD product family	100 μ Ah at 1.5 V	EIA 1812, 4.5 $\times$ 3.2 $\times$ 1.1 mm	Reflowable SMD; several mA pulse capability; oxide electrolyte	Excellent for RTC, backup, low-duty wake buffering; not enough energy alone	38
Ilika Stereax M300	Prototype/ customer-shipment phase; production transfer through Cirtec	300 μ Ah, >1000 cycles, 12 mW pulse power	3.6 $\times$ 5.6 $\times$ 1.1 mm	Ultra-thin solid-state architecture; customer prototype shipments started Dec 2025	Good for strict all-SSB demonstrators and medical-scale devices; still too small alone for all-day camera use	39
ITEN Powency PWY0150S / PWY0250S	Production launch announced; 150 μ Ah in pre-production, 250 μ Ah as engineering sample	150 μ Ah / 250 μ Ah	25 mm ³ SMD package, 73 mg	High-power density; >100C discharge on family page; recharge to 80% in under 15 min on family page	Strong burst-power buffer; still microbattery-scale energy today	40
ITEN roadmap families	Product page publishes roadmap	Powency family up to 10 mAh; Flowency targeted at 10s of mAh to a few Ah for wearables	Optimized SMD today; wearables roadmap for larger devices	Public roadmap implies move from burst microcells toward wearable runtime cells	Most interesting near-/mid-term wearable SSB path if it materializes on schedule	41

Supplier and product	Public status	Publicly stated capacity or density	Form factor	Charging / safety note	Suitability for pendant	Evidence
ProLogium ceramic battery	Pilot / automotive-scale public performance claims	811.6–860.6 Wh/L and 356.3–359.2 Wh/kg; 5–80% charge in 6.4 min on current Gen 4 page	Larger-format cells, not pendant-targeted SKUs	Ceramic design positions safety and fast charge as core value	Strong evidence that density is not the long-term blocker; weak evidence of immediate small-pack availability	42
TDK next-generation material	R&D announcement	1,000 Wh/L material target for next-gen solid-state battery	Not yet a public pendant-ready packaged SKU	Oxide electrolyte and lithium-alloy anodes; explicitly aimed at wearables	Important strategic signal, but not a current product basis	43

The practical implication is stark. If one assumes a mixed-use target of roughly **12 hours at 20 mW average**, the device needs about **240 mWh**. At **800–1,000 Wh/L**, that is only around **0.24–0.30 cm³** of active-cell volume before packaging overhead, so future ceramic cells could absolutely fit inside a pendant. But today's TDK CeraCharge cell stores only about **0.15 mWh** from its public 100 µAh and 1.5 V rating, which means a 240 mWh target would take on the order of **1,600 cells**. That is why the real product question is not “can solid-state chemistry in principle fit,” but “can a sourceable packaged SSB for this form factor exist on the schedule you need.” 44

For charging and safety, the cleanest implementation is still a **single-pack architecture** with a strong PMIC, pack NTC, suspend-on-thermal-rise policy, and product-level current limiting. Series-stacking many microcells is architecturally possible, but it multiplies balancing, assembly, yield, and qualification complexity. The product should be designed around **IEC 62133-2** for lithium portable batteries, product-level **IEC/UL 62368-1** safety, **ISO 10993** for prolonged skin-contact material decisions, and **UN 38.3 / IATA** transport requirements for prototypes and production packs. 45

Power budget and battery-life estimate

The table below is a **first-order engineering estimate**, anchored to public sensor/NPU numbers and the assumptions in this report. It is not a substitute for oscilloscope-level power tracing.

Workload	Public component anchors	Estimated device average power	Runtime on a future 150 mWh pack	Comment
Deep idle	BLE control SoC, RTC, button wake, IMU asleep	1–2 mW	75–150 h	Requires aggressive radio duty-cycling and no active camera
Always-on contextual wake	WiseEye2 public 1.2 mW example + HM0360/HM01B0 sub-mW to low-mW wake sensing + light controller overhead	3–6 mW	25–50 h	Feasible with future wearable SSB; not with today's microbattery parts alone
Explicit snapshot and classify	OV2740 90 mW active + RT700 NPU roughly 60–71 mW during inference + controller/storage overhead	160–220 mW while active	0.7–0.9 h if sustained continuously	Real use case is burst-only: seconds at a time
Audio-first context with rare image bursts	NDP120 wake audio + control MCU, camera mostly off	10–30 mW	5–15 h	This is the easiest path to acceptable runtime
Live preview / Wi-Fi-heavy upload	Main camera + main compute + radio active	400–800 mW	0.2–0.4 h	Poor fit for pendant thermals and current SSB availability

Evidence anchors: WiseEye2 1.2 mW average in a public VGA 1 FPS person-detection example; HM01B0 under 2 mW at QVGA 30 FPS; OV2740 90 mW active; RT700 NPU about 59–71 mW for published MLPerf Tiny inference at 325 MHz; NDP120 35.29 μ J/inference at a low-energy benchmark point. ⁴⁶

A current TDK CeraCharge-class cell would be exhausted in **minutes**, not hours, even in the low-power modes above. By contrast, a future **120–240 mWh** ceramic pack would materially change feasibility for episodic use, especially if the camera is dark most of the time and the wake path stays below about **5 mW**.

Thermally, using the assumed **35 × 30 × 8 mm** envelope yields a total external area of roughly **31 cm²**. A first-order free-air estimate using natural convection plus radiation gives an effective thermal resistance on the order of **23–32 K/W**; with skin contact, lanyard obstruction, and imperfect spreading, the effective resistance can easily move toward **40–50 K/W**. That implies a rough shell rise of about **6–8°C at 250 mW** in favorable free-air conditions and potentially **10–15°C or more when worn**. The design consequence is simple: keep the average shell load low, keep hot bursts short, spread heat into a graphite or thin copper plane, and never combine charging with heavy AI/camera activity unless the enclosure temperature margin is wide.

Ternary computing and software strategy

The most credible ternary path for this pendant is algorithmic, not transistor-native. Recent work from Microsoft ⁴⁷ on BitNet b1.58 argues that ternary weights in $\{-1, 0, +1\}$ can preserve strong model quality while dramatically reducing storage and arithmetic cost. bitnet.cpp then pushes that idea into practical edge inference with specialized CPU kernels. In parallel, xTern shows how ternary neural-network inference can be accelerated on general-purpose cores through a lightweight ISA extension, while recent ternary MAC and logic-in-memory papers continue to show area and power advantages for MVL circuits. The only public “commercial ternary processor” I found is a research-oriented board from Ternary Computer System ⁴⁸, which is interesting but not a volume-proven wearable platform. ⁴⁹

Ternary implementation paths

Path	Maturity	What it buys	What breaks or constrains it	Best use	Evidence
Native MVL / ternary hardware	Research to niche-commercial	Maximum conceptual purity; potential area/power gains; direct trit arithmetic	Tooling immaturity, weak supply chain, no mainstream wearable SoC ecosystem	Long-range R&D, FPGA/ASIC exploration	⁵⁰
ISA-assisted ternary kernels	Early research, but realistic	Addition/skip-friendly ternary ops on more normal cores	Requires custom compiler/runtime support or ISA access	RISC-V or custom MCU experiments	⁵¹
Ternary neural nets on binary hardware	Practical now	Big memory reduction; sparsity from zeros; deployable using custom kernels	Commodity NPUs often do not natively accept ternary execution graphs	Near-term product prototypes	⁵²
Ternary training with INT8 shadow deployment	Most practical for shipping	Keep ternary research benefits upstream while using standard vendor NPUs	Dual-model management and accuracy drift between research and production graphs	Shipping product with future ternary option value	⁵³

The decisive constraint is the **toolchain stack**. Sony’s public IMX500 path requires **fully quantized 8-bit** Keras or ONNX models for deployment and focuses on supported single-image perception models. NXP’s Neutron path is a model conversion flow for its NPU. ST’s public tooling centers on STM32Cube.AI and Neural-ART deployment. Ambiq’s neuralSPOT is focused on TFLM and related runtimes, not a public ternary

execution target. That means a product team should **not** hinge v1 on the hope that a mainstream edge NPU will accept balanced-ternary arithmetic as-is. ⁵⁴

The right software strategy is therefore:

- **Train ternary-weight networks** for always-on wake models and compact context models. Use quantization-aware training, straight-through estimators, and learned thresholds rather than crude post-training clipping. Recent BitNet and small-model ternary work strongly supports this choice. ⁵⁵
- Keep **input stem, first/last layers, fragile normalization paths, and any softmax-heavy blocks in 8-bit or higher precision**. This mixed-precision pattern is usually where the accuracy/energy trade is best.
- Use **ternary weights + INT8 activations** as the default compromise for vision and audio models below roughly the low-tens-of-megabytes scale.
- Where the target chip has a conventional NPU only, export an **INT8 shadow model** for production while retaining the ternary model as the research ground truth.
- If the team wants real ternary execution rather than ternary representation only, deploy on a **programmable CPU/DSP path** first, not on a locked NPU.

For algorithms, the most promising pendant workloads are not LLM-first. They are compact **wake models, scene classifiers, object detectors, text-region detectors, memorability ranking, and audio keyword or voice activity models**. Ternary representation helps most when memory bandwidth dominates, weights are frequently reused, and zeros can be exploited as true skips. That is much more typical of tiny CNNs and compact transformer blocks than of raw video encode/decode paths.

The bottom line is that a ternary pendant should be designed around a **ternary software pipeline with binary-compatible deployment escapes**. That is the only path that is simultaneously rigorous, sourceable, and schedule-aware.

Mechanical design, privacy, security, and regulation

Mechanically, a pendant is governed by **swing, skin, optics, and ingress**. A plausible chest-worn envelope is the assumed **35 × 30 × 8 mm** range with a target mass under about **18 g**. Within that space, the front camera must coexist with lens cover glass, microphone ports, haptic cavity, battery volume, antenna keep-out, and a right-side physical button that is easy to find blindly. If the device is truly pendant-height, a **wide lens** is usually safer for framing than a narrow selfie lens, but that widens the privacy footprint and increases distortion. The IMX708 wide configuration and OV2740-class compact front-camera form factors illustrate the relevant trade. ²¹

Material strategy should separate **structural, RF, and skin-contact** functions. The cleanest stack is a thin metal outer shell or frame for rigidity and heat spreading, a polymer RF window or inner chassis for antennas, a recessed camera window with hard coat or sapphire-like cover, and a sealed side-button plunger with gasket support. For a camera wearable, IP54 is a realistic baseline and IP67 is possible but expensive in volume, especially once microphone vents, sound ports, and connectors are counted. If a pogo-charge design is adopted, sealing becomes much easier than with direct USB-C.

For prolonged skin contact, use materials and coatings that are validated for wearable exposure, not just electronics enclosure appearance. Public wearable-safety guidance from UL points product teams toward **IEC/UL 62368-1** for nonmedical wearables and **ISO 10993** for biocompatibility, and large wearable vendors publicly note special attention to materials that stay in prolonged contact with skin. ⁵⁶

From a privacy and security standpoint, the most important architecture decision is to make the product **metadata-first** and **event-driven**. On-sensor or near-sensor inference reduces the need to store or transmit raw frames, which is both a power advantage and a privacy advantage. Public Sony material for IMX500 explicitly emphasizes lower bandwidth, lower power, and privacy benefits from on-sensor processing. That principle should generalize across the pendant even if IMX500 is not the selected camera. ⁵⁷

For governance and compliance, the relevant public frameworks come from the Federal Communications Commission ⁵⁸, the European Commission ⁵⁹, the European Data Protection Board ⁶⁰, the National Institute of Standards and Technology ⁶¹, ETSI ⁶², UL Solutions ⁶³, and IATA ⁶⁴. FCC approval is required before RF devices are marketed in the U.S.; the EU's RED framework adds cybersecurity obligations for wireless devices; ETSI EN 303 645 remains the clearest public baseline for consumer-IoT security controls; NIST wearable guidance reinforces secure boot, authentication, and protected update/data paths; the EDPB's privacy-by-design and video-device guidance highlights the heightened risks of video and biometric processing; ICO guidance explicitly recommends encryption for wireless video links; UL points to 62368-1, 62133, and ISO 10993 as foundational wearable requirements; and prototype and production batteries still must clear transport rules such as UN 38.3. ⁶⁵

That regulatory posture leads to concrete product rules:

- **Visible recording cue:** dedicated LED near the camera, not hidden in a side seam.
- **Silent but obvious acknowledgement:** haptic pulse on start/stop.
- **Hardware privacy hold:** long-press on the right-side button disables capture path until released or toggled back.
- **Security baseline:** secure boot, signed OTA, per-device keys, encrypted local storage, BLE pairing hardening, and auditable deletion behavior.
- **Data minimization:** no face recognition by default, no indefinite raw-frame cache, region-aware policy settings for audio capture.
- **Market positioning:** avoid medical claims in v1 unless prepared for a different regulatory regime.

Manufacturing, cost, roadmap, and testing

Manufacturability is materially easier if the first hardware revision is **modular in function but monolithic in assembly**: one rigid-flex main body, one camera interposer, one battery cavity, and a production test interface that never appears on the final industrial design. The biggest DFM risks are not the microphones or the button. They are **camera alignment**, **antenna detune on body**, **solid-state battery sourcing**, and the interaction between heat spreading and skin comfort.

Supplier and supply-chain comparison

Subsystem	Near-term supplier posture	Why this is attractive	Main risk	Evidence
Main compact camera	OmniVision OV2740-class or Sony IMX708-class ecosystem	Compact form factor, public module path, strong image specs for front-facing capture	Different trade between autofocus/ photo quality and low-power burst capture	66
Always-on wake vision	Himax HM01B0/ HM0360/WiseEye2	Best public ultra-low-power vision story for wearables and wake sensing	Lower “hero image” quality; may require second camera if rich capture matters	67
Main MCU/ NPU	NXP RT700 or ST STM32N6	Strong public tooling and embedded-AI documentation	Both are still conventional low-bit pipelines, not native ternary platforms	68
Control/ radio	Ambiq Apollo510B or Nordic nRF54L15	Low power, modern BLE, wearable-friendly memory/power envelopes	Limited headroom for richer vision unless paired with a second compute device	69
Audio wake sidecar	Syntiant NDP120	Mature always-on audio specialization	Adds BOM and software orchestration complexity	27
Battery now	TDK / Iluka / ITEN	Real public microbattery products or production launches exist	Energy capacity remains the gating issue	70
Battery future	ProLogium and next-gen TDK material path	Public density trajectory is strong enough for true pendant packs	Schedule and package availability are not yet matched to this exact product class	71

The principal supply-chain conclusion is that **camera and compute are sourceable; solid-state energy is the schedule risk**. If time-to-market matters more than strict chemistry purity, the electronics should be designed so the battery compartment and PMIC can accept a future higher-capacity ceramic pack without re-architecting the whole product.

BOM scenarios

The ranges below are **directional engineering estimates**, not supplier quotes. They are inferred from the sourced component classes above, public module/list information where available, and typical wearable manufacturing multipliers. They exclude NRE, tooling, test-fixture development, certification lab spend, and app/cloud work.

Scenario	Architecture summary	Expected runtime profile	Approx. BOM at 1k	Approx. BOM at 10k	What it is good for
Strict all-solid-state demonstrator	HM01B0/HM0360 or HM11B1 wake vision, WiseEye2-class wake compute, BLE control, mics, haptics, stacked micro-SSB	Minutes to low-hours depending on trigger duty cycle; not all-day	\$70-\$110	\$35-\$55	Research, academic demos, privacy experiments, ternary wake-pipeline validation
Balanced research prototype	OV2740 main camera, RT700-class main MCU/NPU, nRF54L15 or Apollo510B control plane, NDP120 optional, micro-SSB buffer or early larger SSB	Roughly 1-4 hours mixed episodic usage if battery is still micro/meso-scale	\$90-\$140	\$45-\$75	Best platform for serious system exploration
Commercializable SSB-ready architecture	Same electronics, enclosure and PMIC sized for a future 120-240 mWh ceramic pack	Roughly 6-12 hours mixed episodic use once a real pendant-class SSB exists	\$95-\$150 initially; future cell pricing uncertain	\$55-\$95 plus future cell cost curve	Best path if the goal is a real product, not just a science project

If a **custom ternary ASIC** is seriously considered, the economics change completely. The silicon program itself becomes the project. For a pendant startup or new product line, that is rarely justified before the product proves demand, capture UX, and privacy trust.

Development roadmap

A realistic program should prove **power policy and privacy UX before miniaturization**. The wrong order is to miniaturize first and then find that the product is too warm, too battery-limited, or too ambiguous in public use.

```

gantt
  title Development roadmap
  dateFormat YYYY-MM-DD
  axisFormat %b %Y

```

section Discovery	
Requirements and risk freeze	:a1, 2026-05-15, 30d
Power and privacy architecture	:a2, after a1, 45d
section Bring-up	
Dev-board mule and data capture	:b1, 2026-07-01, 45d
Wake-model and ternary pipeline	:b2, after b1, 45d
section EVT	
Split-board pendant EVT	:c1, 2026-09-15, 60d
Thermal and battery iteration	:c2, after c1, 45d
section DVT	
Rigid-flex EVT2 and optics tuning	:d1, 2026-12-01, 60d
RF, EMC, ingress pre-scan	:d2, 2027-01-01, 45d
Privacy and security red-team	:d3, 2027-01-15, 45d
section PVT	
Pilot tooling and test fixtures	:e1, 2027-03-01, 45d
Certification and field trial	:e2, after e1, 60d

Prototyping steps

The fastest rigorous prototyping sequence is:

- **Bench mule:** separate dev boards for the wake path, main camera, main compute, and radio. The goal is to collect real power traces and trigger statistics, not to look wearable.
- **Wearable split-board EVT:** a mechanically representative enclosure with a rigid-flex harness, real button placement, real microphones, and a thermal spreader. The goal is comfort, optics, and heat.
- **Battery-specific EVT2:** once the power policy is stable, integrate the chosen solid-state pack candidate and charger/PMIC path. The goal is leakage, suspend behavior, charging thermals, and loss-of-power robustness.
- **SSB-ready PVT candidate:** freeze the enclosure, test fixtures, pogo alignment, camera calibration flow, and secure provisioning line.

Testing plan

The testing plan should be broader than ordinary small-electronics EVT because this product sits at the intersection of **camera**, **wearable**, **wireless**, and **privacy-heavy** behavior.

Electrical and power

- Rail-by-rail current profiling across every state transition.
- Button-to-capture latency, wake latency, haptic latency.
- Deep-sleep retention, battery leakage, charger suspend/resume behavior.
- Brownout and file-system robustness during unexpected removal.

Optics, acoustics, and thermal

- Camera IQ, MTF, flare, wide-angle distortion, low-light cadence.
- Mic sensitivity spread, clothing rustle rejection, speech intelligibility on chest.
- Surface-temperature mapping on neck/chest phantoms during capture and charging.
- Lens-window fog, sweat, and contamination testing.

Mechanical and environmental

- Drop, tumble, torsion, lanyard pull, button life-cycle, and vent clog tests.
- Sweat, cosmetic corrosion, ingress, thermal cycling, and adhesive aging.
- Antenna performance on-body, off-body, and with different clothing materials.

Security, privacy, and compliance

- Secure boot, OTA rollback resistance, key injection, pairing attacks.
- LED/haptic signaling usability studies to ensure users and bystanders understand capture state.
- Retention/delete verification and incident logging without privacy over-collection.
- Formal pre-compliance for FCC/RED/EMC plus battery shipping, safety, and wearable exposure pathways. ⁷²

The strategic recommendation is therefore clear: **build the first pendant as an SSB-ready, ternary-software-first, episodic-capture system**. Use present-day solid-state cells for demonstrators and wake-heavy prototypes, but do not pretend that today's publicly available micro-SSBs are enough for an all-day camera pendant. On the compute side, treat ternary as a representation and systems strategy now, and as native silicon only later, if the product proves it deserves a custom chip.

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